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cat << 'EOF' > README.md
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# EduSphere-API: Relational MongoDB College Management Backend
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A high-performance, purely backend REST API for a **College Management System** built using **Java (Spring Boot v3.x)** and **MongoDB**.

This system acts as a headless administrative backend that models, links, and manages data across four core college entities: **Staff, Students, Books, and Departments**. It completely replaces traditional SQL tables with scalable NoSQL collections while maintaining logical, real-world data relationships using Object-Document Mapping (ODM).

🎯 Project Objectives

Decoupled Backend Architecture: A zero-frontend, headless system built exclusively for REST API validation using **Postman** and **Swagger UI**.

NoSQL Data Modeling & Relationship Mapping: Uses MongoDB document referencing (`@DocumentReference`) to successfully establish relational entity constraints in a non-relational database.

Extended Administrative CRUD: Implements deep data operations including targeted record insertion, multi-entity updates, exact-match deletions, and structural collection wiping (`Drop`).

Self-Documenting Codebase: Integrates native OpenAPI specifications to offer an interactive local API testing playground out-of-the-box.

🛠️ Tech Stack & Frameworks

- * **Language & Runtime:** Java 17+ / Maven
- * **Core Framework:** Spring Boot v3.x
- * **Data Access Layer:** Spring Data MongoDB (Object-Document Mapper)
- * **API Testing & Docs:** Springdoc OpenAPI v2 (Swagger UI) & Postman
- * **Database:** MongoDB (Local Community Server or Atlas Cloud instance)
- * **Boilerplate Reduction:** Lombok

🏗️ Architectural Class Blueprint

The project implements a clean **Layered Architecture** (``Controller`` -> ``Repository`` -> ``Database Document Model``).

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college-management-system/

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|— src/main/java/com/college/cms/

| |— CollegeManagementSystemApplication.java # Main entry point

| |

| |— model/ # Data Entities / Documents

| | |— Department.java

| | |— Staff.java

| | |— Student.java

| | |— Book.java

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| |
| |─ repository/                # Data Access Layer (CRUD Core)
| |   └─ DepartmentRepository.java
| |   └─ StaffRepository.java
| |   └─ StudentRepository.java
| |   └─ BookRepository.java
| |
| |─ controller/                # REST API Endpoints Layer
| |   └─ DepartmentController.java
| |   └─ StaffController.java
| |   └─ StudentController.java
| |   └─ BookController.java
|
└─ src/main/resources/
    └─ application.properties    # MongoDB & Server Properties
```